

MA.6.NSO.3.3

Benchmark

MA.6.NSO.3.3 Evaluate positive rational numbers and integers with natural number exponents.

Benchmark Clarifications:

Clarification 1: Within this benchmark, expectations include using natural number exponents up to 5.

Connecting Benchmarks/Horizontal Alignment

- MA.6.NSO.2.1, MA.6.NSO.2.2
- MA.6.NSO.3.4
- MA.6.NSO.4.2
- MA.6.AR.1.3
- MA.6.GR.2.3

Terms from the K-12 Glossary

- Base (of an exponent)
- Expression
- Exponents
- Factors
- Integers
- Rational Number
- Natural Number

Vertical Alignment

Previous Benchmarks

- MA.5.NSO.2.1, MA.5.NSO.2.4
- MA.5.FR.2.2

Next Benchmarks

- MA.7.NSO.1.1
- MA.7.NSO.2.1

Purpose and Instructional Strategies

In grade 5, students multiplied multi-digit whole numbers and fractions. In the grade 6, students evaluate positive rational numbers and integers with natural number exponents. In grade 7, students will apply the Laws of Exponents with rational number bases and whole-number exponents.

- Instruction focuses on the connection to repeated multiplication.
- Instruction allows student flexibility in their solution (*MTR.2.1*).
 - For example, allow for both fraction and decimal response when the base is a fraction or a decimal.
- Instruction provides opportunities for students to continue to practice and apply multiplying fractions by fractions, decimals by decimals, and integers by integers (*MTR.5.1*).
- Instruction includes the use of technology to explore and evaluate positive rational numbers and integers with natural number exponents.
- The expectation of this benchmark is the application of exponents, not determining the value in expressions with multiple operations.

- For example, $(-12)^4$ and $\left(\frac{21}{5}\right)^3$ would be appropriate, but $1.258^3 - 12^2$ or $(215^3)(71^2)$ would not be appropriate.

Common Misconceptions or Errors

- Students may multiply the base by the exponent instead of multiplying the base by itself as many times as directed by the exponent. Instruction includes students using expanded form to represent multiplication.
 - For example, 1.258^3 written in expanded form is $(1.258)(1.258)(1.258)$.
- Some students may incorrectly apply or use incorrect notation when exponents are applied to negative integers. If a negative integer has an exponent, the negative base must be in parentheses and the exponent is on the outside of the parentheses (*MTR.5.1*).
 - For example, $(-12)^4 = (-12)(-12)(-12)(-12)$ is not the same as $-12^4 = -(12)(12)(12)(12)$.

Strategies to Support Tiered Instruction

- Instruction includes rewriting exponential expressions in expanded form to represent the multiplication before evaluating.
 - For example:
 - $\left(\frac{1}{4}\right)^3 = \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1 \cdot 1 \cdot 1}{4 \cdot 4 \cdot 4} = \frac{1}{64}$
 - $(-6)^3 = (-6)(-6)(-6) = 36(-6) = -216$
 - $1.258^3 = (1.258)(1.258)(1.258)$
- Teacher creates and posts an anchor chart with visual representations of the base of an exponent, exponent, power, and factor. Teacher encourages students to utilize the anchor chart to assist in correct academic vocabulary when evaluating exponential expressions.
- Instruction includes the use of algebra tiles to represent and evaluate numerical exponential expressions.
- Teacher provides students with flashcards to practice and reinforce academic vocabulary.

Instructional Tasks

Instructional Task 1 (MTR.4.1)

Robin determines the volume of a cube with side lengths of 3.4 cm to be 10.2 cm^3 . Mickey says the volume is 39.304 cm^3 . Which person is correct and why?

Instructional Task 2 (MTR.4.1)

Sean told Parker that $-5^4 = (-5)^4$. Parker told Sean that he is incorrect. Is Sean correct or is Parker correct? How do you know?

Instructional Task 3 (MTR.6.1)

Determine if 8^4 is equivalent to 4^8 . Explain your reasoning.

Instructional Items

Instructional Item 1

What is the value of the expression $\left(\frac{1}{3}\right)^3$?

Instructional Item 2

What is the value of the expression $(-3)^4$?

Instructional Item 3

What is the value of the expression 0.1^3 ?

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*